Forecasting Hourly Bike Rentals
Micro-Project 4

https://github.com/mascon0423/ANA500/tree/main/week4

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Problem Statement

- Bike rental activity changes throughout the day. Operators need a useful estimate of the next hour's system-wide activity.
- Question: How accurately can the previous 24 hours of rentals and known calendar information forecast the next hour's rentals?
- Week 4 uses hourly bike-sharing data so the RNN learns from a meaningful time sequence. The Adult dataset used earlier describes different people.
- The outcome is a numerical rental count. Recorded rentals do not reveal unmet demand or individual station needs.

Hypothesis Formulation

- A deep neural network using recent activity and calendar patterns will reduce forecast error compared with repeating the previous hour or the same hour yesterday.
- Possible explanation: Regular schedules and recent rental activity make some short-term patterns predictable.
- Select the preferred model and its training checkpoint using validation RMSE. Then evaluate the fixed models on later test hours.
- Improved forecasts support a predictive relationship in this sample. They do not establish causation or guarantee future performance.

Acquire

- Source: UCI Bike Sharing, Capital Bikeshare in Washington, D.C., during 2011-2012.
- 17,379 recorded hours and 17 columns in hour.csv. The CSV and source README agree on the row count.
- The notebook includes the unchanged CSV, source documentation, attribution, and a checksum that verifies the input.
- Tools used: Jupyter Notebook, Python, Pandas, NumPy, Matplotlib, and PyTorch 2.6.0. No API key is required.
- Fanaee-T (2013). Bike Sharing. https://doi.org/10.24432/C5W894

Prepare: Histories and Inputs

- 165 clock hours have no source row. Keep them missing instead of replacing them with zero.
- Use 15,657 target hours with a complete preceding 24 hours. Exclude 1,722 recorded targets that lack sufficient history.
- Inputs: 24 past rental counts, sine/cosine pairs for hour, weekday, and annual position, plus the target hour's working-day flag.
- Exclude target-hour weather, casual counts, registered counts, and the target count. Casual plus registered equals the outcome.
- Normalize counts with 2011 data only. Preserve chronological order within every input sequence.

Prepare: Chronological Evaluation

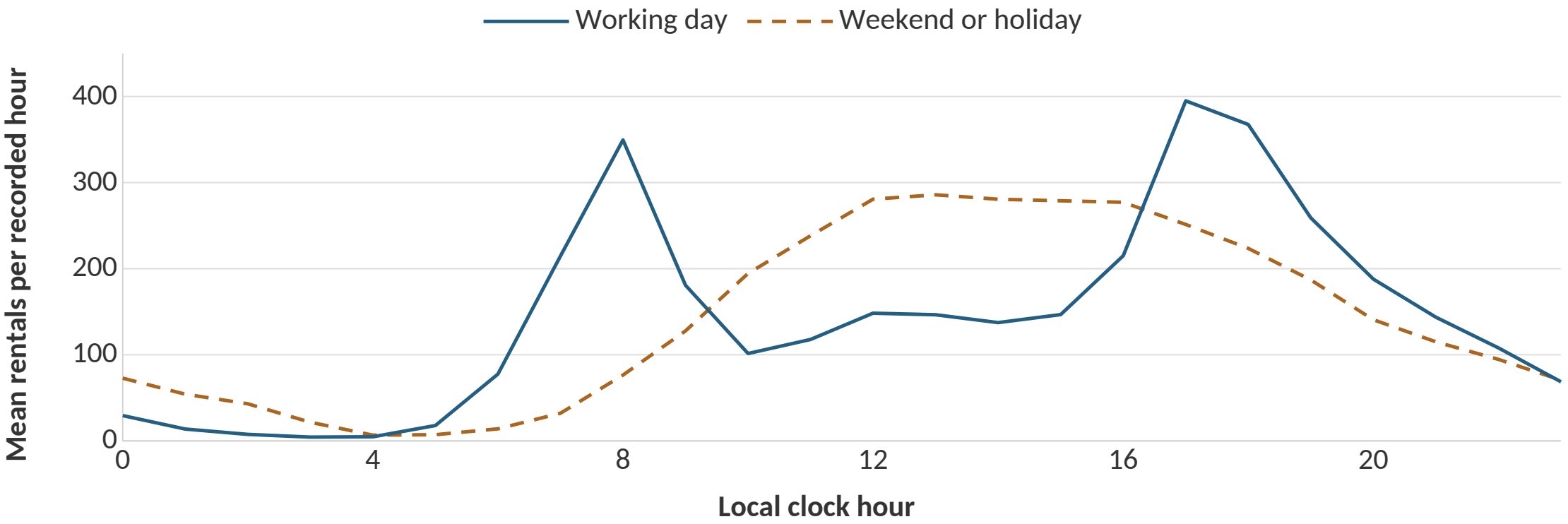
Split	Target period	Eligible hours	Purpose
Training	2011	7,279	Learn weights
Validation	Jan-Jun 2012	4,120	Choose model and epoch
Test	Jul-Dec 2012	4,258	Evaluate fixed models

Earlier observed hours may cross split boundaries as history. Future hours never enter an input.

Forecast one hour ahead, observe the actual count, then forecast the next hour. Model weights stay fixed during testing.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Prepare: Daily Rental Patterns



In 2011, working-day activity has morning and evening peaks. These training-period patterns motivate the calendar inputs.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

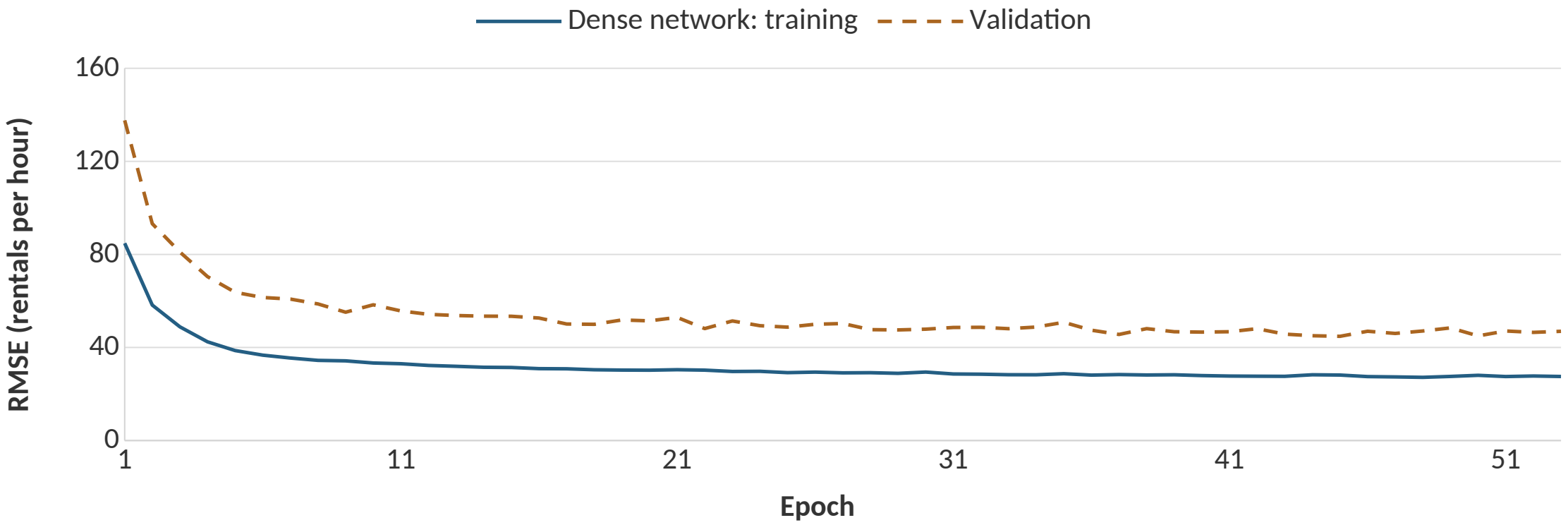
Analyze: Deep Learning Models

Model	Hidden layers	How it uses history	Parameters
Dense network	64, then 32 units	Reads all 24 lags together	4,161
Basic RNN	2 recurrent layers of 32; dense 16	Updates a hidden state in order	3,889
LSTM	2 recurrent layers of 32; dense 16	An RNN with gated memory	13,585

All neural models use the same information and output one rental count. Two simple baselines use the previous hour or the same hour yesterday.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Analyze: Training and Checkpoints



Adam, learning rate 0.001, batches of 256, at most 60 epochs. Restore the checkpoint with the lowest validation RMSE.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Analyze: Validation Results

Model	Validation RMSE	Best epoch	Epochs run
Previous hour	113.42	N/A	N/A
Same hour yesterday	123.61	N/A	N/A
Dense network	44.78	45	53
Basic RNN	59.58	45	53
LSTM	64.09	33	41

Dense network has the lowest validation RMSE. Selection is fixed before test scoring. One seed and one configuration per architecture.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Report: Test Results

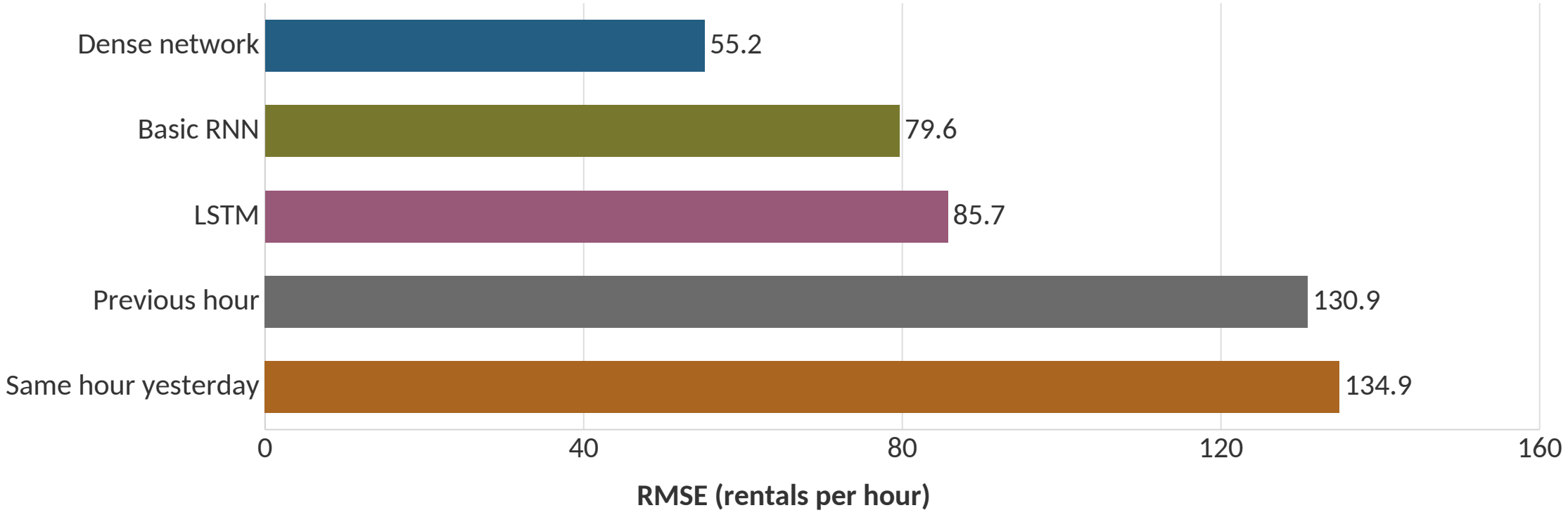
Model	MAE	RMSE	R-squared
Previous hour	86.91	130.90	0.647
Same hour yesterday	80.94	134.87	0.625
Dense network	35.61	55.18	0.937
Basic RNN	48.60	79.65	0.869
LSTM	51.30	85.70	0.849

MAE and RMSE are in rentals per hour. Test period: July-December 2012.

Same 4,258 eligible test hours for all models. Lower MAE/RMSE is better. The validation choice remains dense network.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

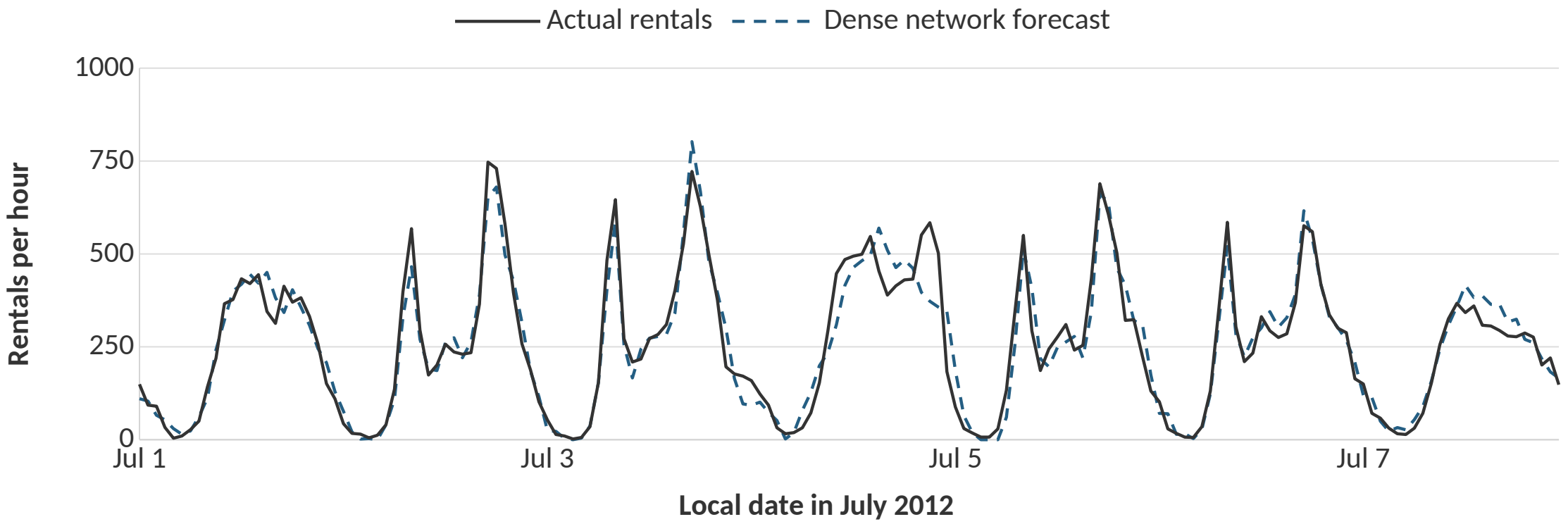
Report: Comparing Forecast Errors



Dense network reduces test RMSE by 57.8% versus the validation-selected baseline (previous hour).

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

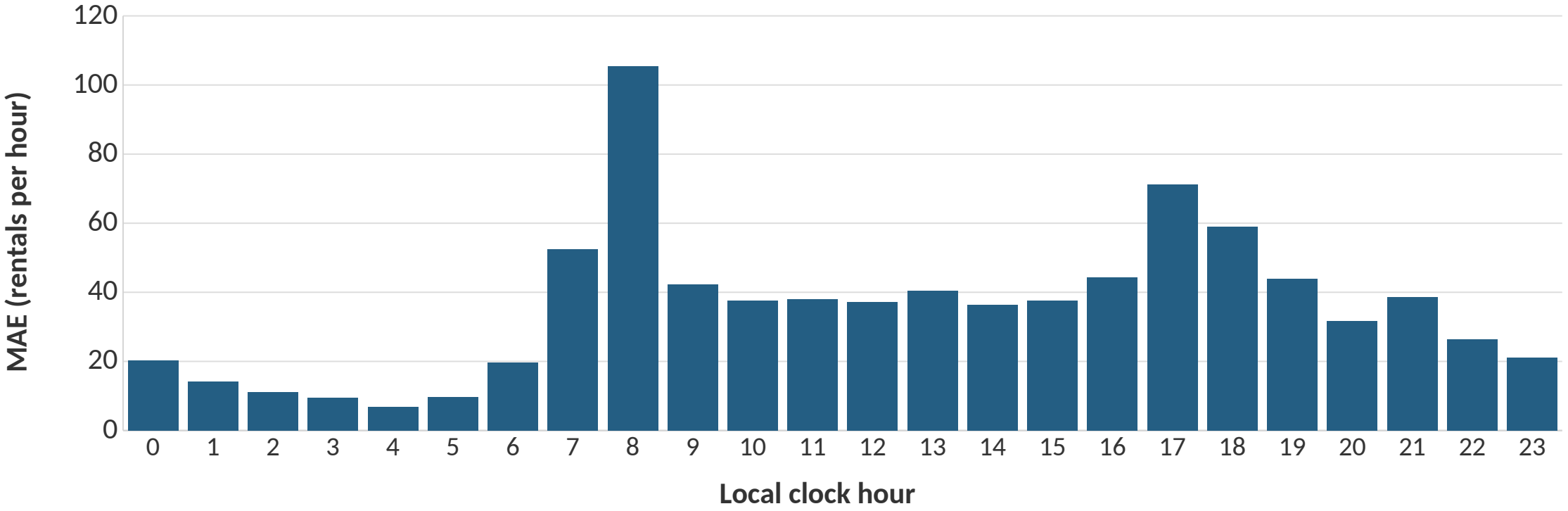
Report: First Test Week



July 1-7, 2012 was chosen by date, not forecast accuracy. These are rolling one-hour forecasts using newly observed counts each hour.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Report: Error by Clock Hour



The largest average error is at 08:00: 105.48 rentals (n = 177). Group sizes range from 175 to 178.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Report: Findings and Limitations

- Dense network was selected on validation data and also has the lowest test RMSE (55.18 rentals). Its MAE is 35.61 and R-squared is 0.937.
- The selected model beats both simple baselines on this test period, supporting the predictive hypothesis. The basic RNN and LSTM also improve on the baselines.
- A recurrent architecture did not win this comparison. Different sizes, one random seed, and one time split limit general conclusions.
- Historical data, excluded gap periods, omitted weather, and immediate-count availability limit practical use. Timezone offsets are not supplied.
- Forecasts are point estimates without calibrated uncertainty intervals. They do not measure unmet demand or establish causes.

Act: Apply the Selected Model

Forecast hour	Actual rentals	Forecast rentals
2012-07-01 00:00	149	110.2
2012-07-01 01:00	93	103.9
2012-07-01 02:00	90	65.7

Selected method: Dense network. Average test miss: 35.61 rentals per hour.

First three eligible test hours, in chronological order. Forecasts estimate system-wide activity and do not specify station inventory.

Source: UCI Bike Sharing, 2011-2012. Forecasts use complete 24-hour histories.

Act: Decision and Next Steps

- Use dense network as a candidate for a monitored trial with current data. Keep the simple baselines for comparison.
- Record each forecast before the hour begins. Review weekly errors, agree on acceptable error with the operator, and flag incomplete histories.
- Before operational use, test additional time periods and seeds, reporting delays, and prediction intervals. Station decisions need station and inventory data.
- This completes Step 5, Act, with a practical proposal and application to held-out hours. It is not a live deployment.

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Data and license: https://doi.org/10.24432/C5W894